

Spine RQ3: Realized Work and Dynamic-Update Latency

Correctness-Gated Cycle-Level Evidence

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Abstract

This report tests whether Spine latency follows work realized by the actual execution rather than graph capacity or a fitted edge-count proxy. It covers 54 dual-oracle-admitted executions and all five requested mechanisms: zero-net, shallow insertion, deep carry, residual PageRank correction, and SSSP/CC nonmonotonic fallback. Every plotted latency ledger closes exactly to end-to-end cycles.

1 Representative end-to-end breakdown

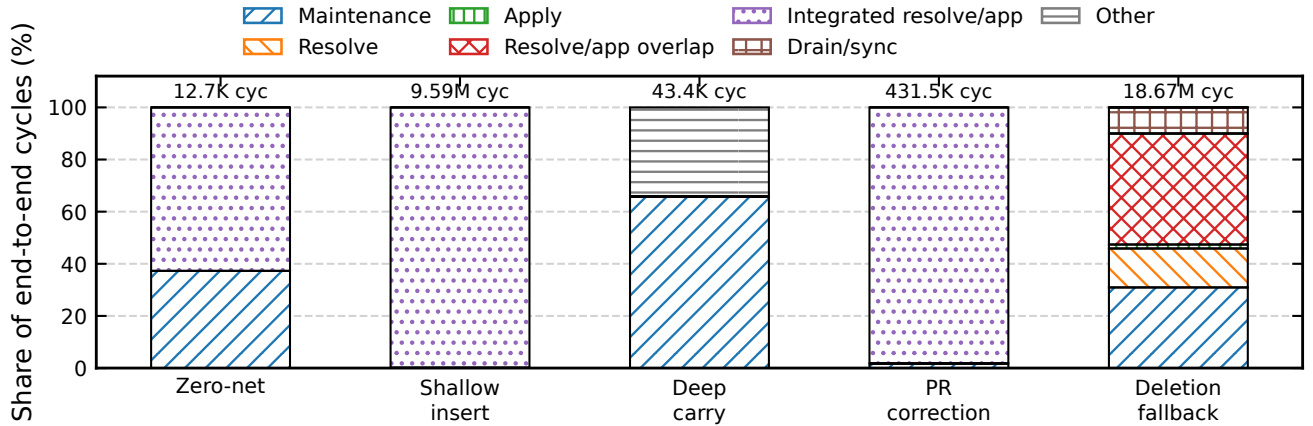


Figure 1: Exclusive, overlap-aware critical-path composition. Absolute device cycles are printed above each normalized bar. Integrated resolve/app is used only when the runner does not expose split timestamps.

Case	Algorithm	E2E cyc	Maint.	Integrated R/A	Sync
Zero-net	CC	12,671	4,720	7,951	0
Shallow insert	CC	9,587,612	7,390	9,580,222	0
Deep carry	SSSP	43,407	28,548	0	0
PR correction	Residual PR	431,479	7,525	423,954	0
Deletion fallback	SSSP	18,668,757	5,773,955	0	1,870,084

Table 1: Absolute timing for the selected representative of each class. Selection is deterministic: the maximum-cycle eligible execution.

2 Realized-work validation

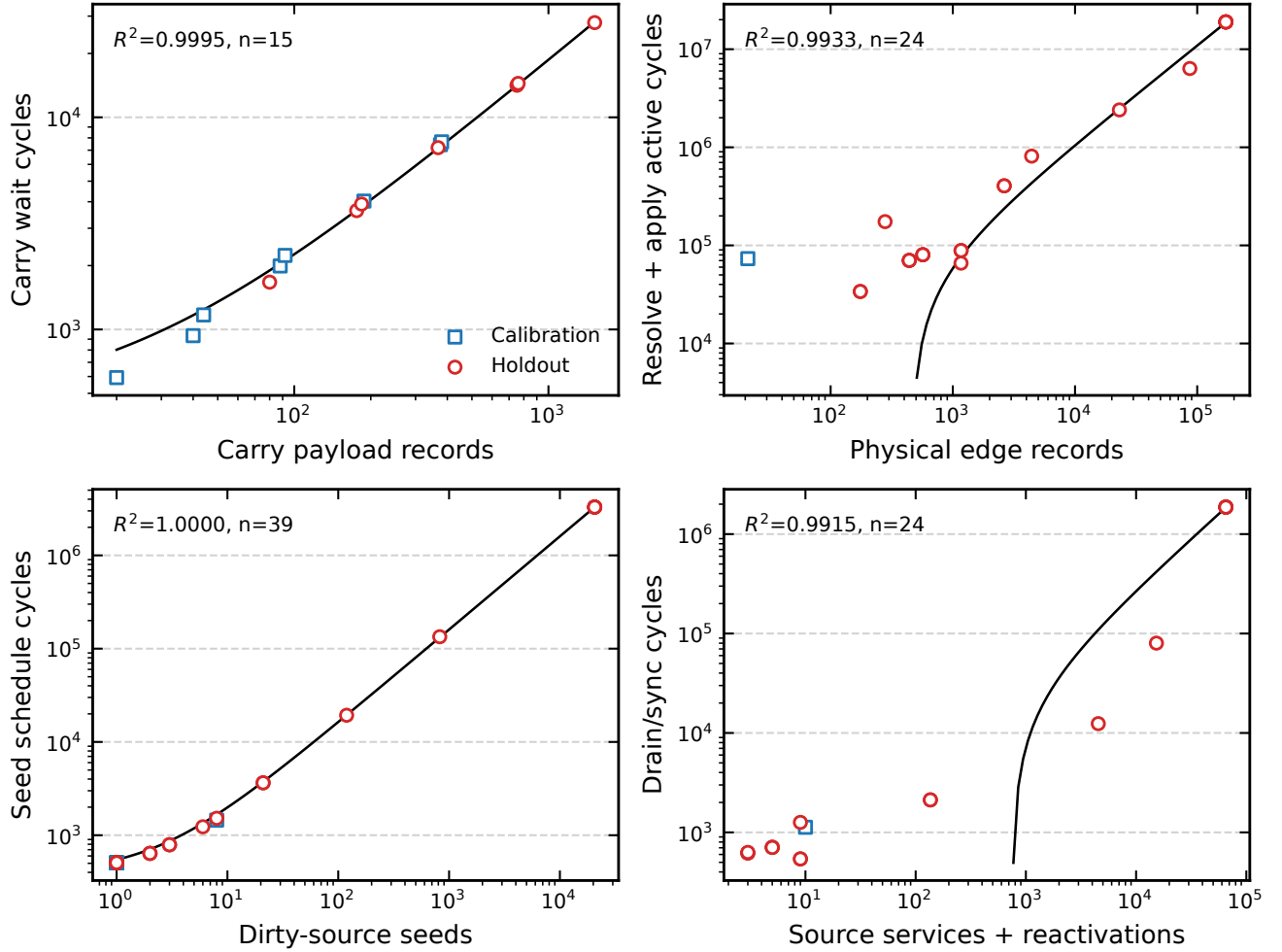


Figure 2: Direct execution counters versus mechanism-local cycles. Blue squares are calibration cases and red circles are holdout traces. Fits use all admitted rows with positive supported work.

Mechanism	Samples	Slope	R^2
carry	15	18.242	0.9995
physical_resolve_apply	24	109.370	0.9933
seed	39	159.929	1.0000
drain_sync	24	28.773	0.9915

Table 2: The carry predictor counts old payload reads, merge inputs, and output rewrites. Cursor bitmap inspection is exported separately.

Interpretation. Carry work, physical resolve/app work, seed work, and drain/synchronization work explain their corresponding cycle counters with R^2 above 0.99 in the covered population. Directory request count ($R^2 = 0.0011$) and the current aggregate switch-work count ($R^2 = 0.0181$) do not transfer across mixed case classes, so no explanatory claim is made for those two mechanisms yet.

Claim boundary. These correlations validate the simulator’s internal cost model and expose bottlenecks. They do not constitute cycle-for-cycle FPGA calibration, total board energy, or proof that one scalar predictor explains every topology.